

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Fieldable MALDI-TOF Bioaerosol Analysis System

Authors: Wayne A. Bryden

Institute: Johns Hopkins University Applied Physics Laboratory

ABSTRACT

The mass spectrometer is known as the most powerful laboratory analytical tool for analysis of a broad spectrum of chemical and biological materials. The applicability of mass spectrometers to field detection problems has been quite limited given the large size, heavy weight, and prohibitive power requirements of the instrumentation. DARPA has funded a program at JHU-APL with the objective of integrating new and emerging mass spectrometer technologies into a highly sensitive, broadband detection system suitable for use in a field setting. The matrix assisted laser desorption/ionization (MALDI) technique has revolutionized biological mass spectrometry. Using this technique in the laboratory, large molecules (particularly proteins) can be employed as signatures of biological materials. These signatures have been used with the proper algorithms to identify microorganisms. The DARPA funded system is being developed to carry this technology into the field in a package with low logistics burden and capable of being operated by lightly trained personnel. The system has demonstrated rapid classification and identification of a variety of bioaerosol components including bacteria (both spores and vegetative cells), viruses and toxins. The biodetection system includes an optimized time-of-flight mass analyzer; an automatic collection system for transport of concentrated airborne particle samples into the mass analyzer; an automated analytical methodology; and automated signal processing, decision aids and data displays. This system is applicable to the rapid (<5minute) detection of airborne microorganisms in a mixed environment.